

Four equal Gaussian cells: covariance deformation and tetrahedral rigidity

mxym/math research project

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Abstract

We prove that a partition of standard Gaussian space into four cells of probability $1/4$ maximizes the sum of squared first moments precisely when it is the central regular tetrahedral partition, extended cylindrically in dimensions above three. Its value is $12(\arctan \sqrt{2})^2/\pi^3$. The argument introduces a trace-normalized Gaussian covariance value function. The Gaussian multi-bubble perimeter theorem puts every full-rank critical value above the tetrahedral value. A constrained mountain-pass argument would contradict any additional maximizer, but its covariance domain has a singular boundary. We handle that boundary by smooth covariance regularization, a normal-cone spectral estimate, and a uniform separation inequality for balanced winning cells. Rank-two and rank-one obstructions then exclude all limiting boundary critical points. The proof is analytic; it uses no numerical search or computer-assisted Gaussian integration. We state the imported Gaussian isoperimetric results explicitly. Partial Lean checks cover finite algebra, not the Gaussian analysis or the complete theorem.

1. The theorem and its scope

Let γ_d be standard Gaussian probability on $\mathbb{R}^d$, with density $(2\pi)^{-d/2}e^{-|x|^2/2}$. A partition is a measurable list $C_1, \dots, C_4$, disjoint and covering space up to null sets. Define

$$m_i = \int_{C_i} x d\gamma_d(x), \quad F(C) = \sum_{i=1}^4 |m_i|^2, \quad \theta = \arctan \sqrt{2}.$$

Theorem 1. If $d \geq 3$ and $\gamma_d(C_i) = 1/4$ for every i , then

$$F(C) \leq F_* := \frac{12\theta^2}{\pi^3}. \tag{1}$$

Equality holds exactly for the following partitions, up to null sets, orthogonal transformations and relabeling. In an orthogonal three-dimensional subspace take

$$\sigma_1 = (1, 1, 1), \quad \sigma_2 = (1, -1, -1), \quad \sigma_3 = (-1, 1, -1), \quad \sigma_4 = (-1, -1, 1),$$

and assign x to the label maximizing $\sigma_i \cdot x$. The cells are central regular tetrahedral cones, with the orthogonal complement unrestricted.

The same bound and equality classification hold for fractional partitions: measurable $0 \leq f_i \leq 1$ with $\sum_i f_i = 1$ almost everywhere and $\int f_i d\gamma_d = 1/4$, using $m_i = \int x f_i d\gamma_d$.

This solves the equal-mass four-cell first-moment question, including Heilman's 2014 Conjecture 3 in dimension three [H14], with regularity added to its stated simplicial-conical classification, and the equal-mass four-cell case of [H19, Conjecture 1.16]. It does not assert the arbitrary-mass conjecture, more than four cells, or noise stability at positive correlation. The arbitrary-mass statement as written in [H19] is false, as the companion note [EX] explains. The literature screen in this package is finite and makes no worldwide priority claim.

2. Balanced Gaussian assignments and their covariance

Write $\mathbf{1} = (1, 1, 1, 1)^T$, $P = I - \mathbf{1}\mathbf{1}^T/4$ and

$$\mathcal{K} = \{Q = Q^T \succeq 0 : Q\mathbf{1} = 0, \text{tr } Q = 1\}, \quad Q_* = P/3.$$

This is a compact convex subset of the six-dimensional Euclidean space of symmetric matrices annihilating $\mathbf{1}$, with the Frobenius inner product. For every such positive semidefinite Q , without the trace restriction, define

$$\mathcal{C}(Q) = \inf_{\lambda \in \mathbb{R}^4} \left\{ \frac{1}{4} \sum_i \lambda_i + \mathbb{E} \max_i (Y_i - \lambda_i) \right\}, \quad Y \sim N(0, Q). \quad (2)$$

Equivalently, it is the largest $\mathbb{E}Y_I$ among couplings with $\mathbb{P}(I = i) = 1/4$. At coincident scores such couplings can split tied labels. Only the dual definition, and the ordinary winning assignment for distinct scores, are needed below.

For completeness, write $Y = MG$ with centered score rows v_i , so $MM^T = Q$. Fix the gauge $\min_i \lambda_i = 0$. The expression in braces, called $\Psi(M, \lambda)$, is at least $\max_i \lambda_i/4$: choose a zero-priced label in the pointwise maximum and use its zero Gaussian mean. Thus a minimizing price exists. For distinct v_i , score ties are null, and differentiation in prices gives

$$\partial_{\lambda_i} \Psi = 1/4 - \gamma(C_i), \quad C_i = \{x : v_i \cdot x - \lambda_i \geq v_j \cdot x - \lambda_j \ \forall j\}. \quad (3)$$

These winning cells therefore have the four required masses. They attain the linear assignment because the winning score is pointwise largest. At degenerate scores the coupling interpretation can also be obtained by approximating Q with positive covariances and taking weakly compact limits of the four fractional assignments on a common Gaussian space. The moment integrals pass to the limit because Gaussian coordinates are in L^2 .

Lemma 2 (continuity and scaling). $\mathcal{C}$ is continuous on the closed covariance cone and $\mathcal{C}(sQ) = \sqrt{s} \mathcal{C}(Q)$ for $s \geq 0$.

Proof. Scaling scores and prices proves homogeneity for $s > 0$, and $\mathcal{C}(0) = 0$. Couple $Y = Q^{1/2}G$ and $\tilde{Y} = \tilde{Q}^{1/2}G$. For every price the difference of expected maxima is at most $\mathbb{E}|(Q^{1/2} - \tilde{Q}^{1/2})G|$.

This bound is independent of prices, hence also bounds the difference of their infima. Continuity of the matrix square root proves the assertion. $\square$

Suppose now that Q is positive definite on $\mathbf{1}^\perp$ and take a centered 4×3 matrix M with $MM^T = Q$. The scores are affinely independent. Every pair has a positive-area winning facet. For its relatively open part Σ_{ij} put

$$A_{ij} = \int_{\Sigma_{ij}} \varphi_3 d\mathcal{H}^2, \quad w_{ij} = \frac{A_{ij}}{|v_i - v_j|}, \quad (Lu)_i = \sum_{j \neq i} w_{ij}(u_i - u_j). \quad (4)$$

At a lower-rank diagram the same definition in ambient dimension three is used, with $A_{ij} = 0$ for an absent facet. Cylindrical coordinates integrate to one. Let B be the matrix of actual cell moments. Gaussian integration by parts gives

$$B = LM, \quad \mathcal{C}(Q) = \langle M, B \rangle = \text{tr}(LQ). \quad (5)$$

Indeed the inward normal to cell i on this facet is $(v_i - v_j)/|v_i - v_j|$. Truncation by large balls justifies integration by parts: their additional Gaussian flux tends to zero exponentially.

Lemma 3 (interior differential identity). In the positive cone $\mathcal{C}$ is smooth, and for every symmetric D with $D\mathbf{1} = 0$,

$$d\mathcal{C}(Q)[D] = \frac{1}{2} \text{tr}(LD). \quad (6)$$

Proof. Smoothness can be seen without differentiating moving corners. The three score differences against label four make a nonsingular affine change of variables; in those variables all four winning regions are fixed polyhedral cones. Gaussian density derivatives in the changed variables are polynomials times locally uniformly dominated Gaussian densities. The price Hessian is L , obtained by normal flux across the facets. It is positive definite in the gauge $\sum \lambda_i = 0$, since the facet graph is complete. The implicit function theorem gives smooth balancing prices. This proves smoothness locally everywhere in the positive cone. Uniqueness of prices in this gauge follows as well: the Hessian is positive definite along any price segment.

Write $D = MAM^T$ with $A = A^T$, and realize $Q + tD$ using $M(t) = M(I + tA)^{1/2}$. The envelope derivative of the balanced score value is B , so

$$d\mathcal{C}(Q)[D] = \frac{1}{2} \text{tr}(B^T MA) = \frac{1}{2} \text{tr}(LD).$$

Every covariance direction has this representation because M maps onto $\mathbf{1}^\perp$. $\square$

3. Separation of the moments of winning cells

Let $\phi(t) = (2\pi)^{-1/2} e^{-t^2/2}$, $q = \Phi^{-1}(3/4)$, $h_0 = \phi(q)$ and $k_0 = 1/(16\phi(0))$. Every cell of mass $1/4$ has moment norm at most h_0 : for a prescribed direction $T = u \cdot X$, let f be any cell indicator (or fractional indicator) of that mass. Pointwise, $(T - q)(f - \mathbf{1}_{T \geq q}) \leq 0$. Integration cancels the q term by equal masses, giving $\int T f \leq \int_{T \geq q} T = \phi(q)$. Consequently, for trace-one scores,

$$0 \leq \mathcal{C}(Q) = \langle M, B \rangle \leq 2h_0. \quad (7)$$

This initially holds in the positive cone, and extends to its boundary by Lemma 2.

Lemma 4 (uniform pair separation). For any distinct-score balanced winning diagram, and every pair $i \neq j$,

$$(b_i - b_j) \cdot \frac{v_i - v_j}{|v_i - v_j|} \geq k_0. \quad (8)$$

Proof. Put $n = (v_i - v_j)/|v_i - v_j|$ and $t = (\lambda_i - \lambda_j)/|v_i - v_j|$. On $D = C_i \cup C_j$ the two labels are separated exactly by $n \cdot X = t$. The subdensity g of $n \cdot X$ restricted to D satisfies $0 \leq g \leq \phi(0)$ and has mass $1/2$. Both sides of t have mass $1/4$. Thus

$$(b_i - b_j) \cdot n = \int |z - t|g(z) dz \geq \int_0^{1/(4\phi(0))} \left(\frac{1}{2} - 2\phi(0)s\right) ds = k_0. \quad (9)$$

Here the constant t cancels by equal masses. The inequality is the layer-cake formula and the bound on the mass in $[t - s, t + s]$. No optimality of the squared moment objective is assumed. $\square$

We will also need the price bound

$$|\lambda_i - \lambda_j| \leq q|v_i - v_j|. \quad (10)$$

Each winning cell lies in its pairwise winning halfspace. Both opposite halfspaces have probability at least $1/4$, giving (10).

4. Boundary continuity of the facet matrix

Lemma 5. Suppose centered score lists $M_n \rightarrow M_0$ in dimension three have pairwise separated rows. Let their balanced prices converge to λ_0 . Then the limiting diagram has four masses $1/4$ and $L_n \rightarrow L_0$, where L_0 is its facet matrix. The assertion allows rank-one or rank-two limits.

Proof. Distinct limiting scores have pairwise tie hyperplanes of Gaussian measure zero. Dominated convergence of the winning indicators gives the four limiting masses. The same dominated convergence applies to first moments, since Gaussian coordinates are integrable.

A codimension-one triple tie is impossible in this positive-mass diagram. Such a tie makes three score vectors affinely collinear with exactly the same affine relation among their prices. One of the three affine scores is then globally a strict convex combination of the other two. Its winning cell is contained in a hyperplane and is null, a contradiction.

For each pair write its facet plane as $n_n \cdot X = t_n$. Its area is $\phi(t_n)$ times the two-dimensional Gaussian probability of the other two winning inequalities restricted to that plane. A continuous local orthonormal frame for $n_n^\perp$ exists near its limiting normal; use it to express these inequalities in fixed two-dimensional Gaussian coordinates. A limiting nonconstant inequality has a Gaussian-null zero set. A constant nonzero one is stable under convergence. A constant zero inequality would be the prohibited codimension-one triple tie. Dominated convergence therefore gives each facet area,

including those that tend to zero. Pair separation keeps its denominator nonzero, proving $L_n \rightarrow L_0$. The limiting flux identity (5) holds by the same integration by parts. $\square$

5. A boundary obstruction for self-moment diagrams

Lemma 6. There is no balanced four-cell winning diagram in dimension three whose four distinct inducing vectors are its own cell moments, whose moment rank is at most two, and whose facet matrix satisfies $L \preceq P$.

We give the complete proof because this is the boundary hypothesis needed later; the lemma does not assume global maximality.

First recall the elementary unrestricted three-cell bound

$$\sum_{i=1}^3 \left| \int_{D_i} x \, d\gamma \right|^2 \leq B_3 := 9/(8\pi). \quad (11)$$

To prove it, if its moment sum is $S > 0$, divide the centered moments by $\sqrt{S}$ to get scores with squared norm sum one. Their span has dimension at most two, and pointwise maximization gives $\sqrt{S} \leq \mathbb{E} \max_i v_i \cdot G$. The winning cells are at most three planar sectors, possibly empty. A sector of angle α has moment norm $\sin(\alpha/2)/\sqrt{2\pi}$. If $x_i = \alpha_i/2$ sum to π , $\sum \sin^2 x_i = 2 + 2 \prod \cos x_i \leq 9/4$: a negative cosine makes the product nonpositive; otherwise concavity of $\log \cos$ bounds the product by $1/8$. Endpoints follow by continuity. Cauchy–Schwarz for the normalized scores and sector moments now gives $\mathbb{E} \max_i v_i \cdot G \leq \sqrt{B_3}$, proving (11). Rank-one cases may be viewed in a plane with an unused coordinate.

The strict profile estimate is

$$h_0^2 > B_3/4 = 9/(32\pi), \quad h_0 > 3\phi(0)/4. \quad (12)$$

Indeed $\pi < 22/7 < 32/9$ gives $\phi(0) > 3/8$ and $e^{-x^2/2} \geq 1 - x^2/2$ gives $\Phi(3/4) > 1/2 + (3/8)(87/128) > 3/4$. Thus $0 < q < 3/4$. Taylor's theorem implies $e^{-q^2} > 1 - 9/16 + (9/16)^2/2 - (9/16)^3/6 = 4637/8192 > 9/16$. This proves both claims in (12).

Now suppose the self-moment rank is two. Write its matrix as M . Flux gives $LM = M$. Hence $W = P - L \succeq 0$ annihilates $\mathbf{1}$ and the two-dimensional column space of M , so

$$W = caa^T, \quad c \geq 0, \quad a \perp \mathbf{1}, \quad a^T M = 0, \quad w_{ij} = 1/4 + ca_i a_j. \quad (13)$$

If all four moments are hull vertices, they form a convex quadrilateral. Its affine dependence has alternating signs, say $a_1, a_3 > 0$ and $a_2, a_4 < 0$. For the scores $s_i = m_i \cdot x - \lambda_i$, the number $K = \sum a_i s_i = -\sum a_i \lambda_i$ is constant. On an ordinary 13 facet, $s_1 = s_3 = T > s_2, s_4$ gives $K = a_2(s_2 - T) + a_4(s_4 - T) > 0$. On a 24 facet it gives $K < 0$. Both diagonal facets cannot exist, but (13) gives both positive weights. This is a contradiction.

Otherwise a moment lies in the hull of the other three. Write $m_4 = \sum_{j=1}^3 \alpha_j m_j$, $\alpha_j \geq 0$, $\sum \alpha_j = 1$. In (13) take $a = (\alpha_1, \alpha_2, \alpha_3, -1)$. Then $w_{4j} \leq 1/4$. Put $t = |m_4|$ and $F = \sum |m_i|^2$. The classical

single-cell Gaussian isoperimetric theorem [B75, ST78] gives the following perimeter lower bound for the fourth cell, which has mass $1/4$:

$$h_0 \leq \sum_j A_{4j} \leq \frac{1}{4} \sum_j |m_4 - m_j| \leq \frac{\sqrt{3}}{4} \sqrt{F + 4t^2}. \quad (14)$$

The squared lengths sum to $F + 4t^2$ because the four moments sum to zero. If $t > 0$, the convex combination gives some $m_4 \cdot m_j \geq t^2$. Merging these two cells and using (11) yields $F + 2t^2 \leq B_3$. Projecting all moments onto m_4/t , two projections are t and $s \geq t$, and the other two sum to $-t - s$. Thus $F \geq t^2 + s^2 + (t + s)^2/2 \geq 4t^2$. For $t = 0$ these same two inequalities hold by merging any other cell. They imply $t^2 \leq B_3/6$ and $F + 4t^2 \leq 4B_3/3$. Equation (14) now gives $h_0 \leq \sqrt{B_3}/2$, contradicting (12). This includes points on hull edges and zero coefficients α_j .

For rank one, the four distinct slopes give four ordered intervals, whose equal masses force endpoints $-q, 0, q$. Their own moments are $(-h_0, h_0 - \phi(0), \phi(0) - h_0, h_0)$. The middle facet has weight

$$w_{23} = \frac{\phi(0)}{2(\phi(0) - h_0)} > 2,$$

by (12). Hence $\text{tr } L \geq 2w_{23} > 4$, contradicting $L \preceq P$ and $\text{tr } P = 3$. Rank zero contradicts distinct moments. This proves Lemma 6. $\square$

6. The regular value and its strict local maximum

Lemma 7. Put $c_* := \sqrt{F_*}$. Then $\mathcal{C}(Q_*) = c_*$, and Q_* is a strict local maximum of $\mathcal{C}$ on $\mathcal{K}$.

We include the local calculation. For regular scores $m_i^0 = a\sigma_i$, the 12 facet is the plane $y + z = 0$ with wedge $X \geq |Y|/\sqrt{2}$. Its angular width is 2θ and Gaussian area is $\phi(0)\theta/\pi$. Its edge length is $2\sqrt{2}a$. Taking $a = \theta/\pi^{3/2}$ makes every weight $1/4$, so flux gives precisely the cell moments m_i^0 . Thus their value is $12a^2 = F_*$. The trace-one scores are m_i^0/c_* , with covariance Q_* , and their assignment value is c_* .

To check strict local maximality, work near the self-moment list and remove rotations by polar decomposition. Perturb the scores by the symmetric map $I + B$, where

$$B = \begin{pmatrix} u & b_z & b_y \\ b_z & v & b_x \\ b_y & b_x & s \end{pmatrix}, \quad c_0 = \frac{\sqrt{2}}{3\theta}, \quad h = \frac{\sqrt{\pi/3}}{\theta}.$$

The latter h is the normalized X -mean of the facet wedge: its mass is θ/π and its unnormalized mean is $\phi(0)\sqrt{2/3}$. A price variation can be parametrized by an apex variation z , with $\delta\lambda_i = m_i^0 \cdot z$ in the zero-sum gauge. Facet flux gives

$$\delta p_i = a\sigma_i \cdot (h(b_x, b_y, b_z) - z).$$

For example the facet conditional mean is he_x on 12. Summing all facet fluxes gives $\delta p_i = (h/(2a))((m_i^0)^T B m_i^0 - a^2 \text{tr } B) - m_i^0 \cdot z$, which is the displayed expression. Balanced prices therefore give $z = h(b_x, b_y, b_z)$.

Set $K_x = c_0(u - (v + s)/2) - (v + s)/2$, with cyclic definitions for K_y, K_z . The two opposite facet-weight derivatives are

$$4\delta w_{12} = K_x + (c_0 - 1)b_x - hz_x, \quad 4\delta w_{34} = K_x - (c_0 - 1)b_x + hz_x. \quad (15)$$

Here a diagonal perturbation changes the half wedge angle by $(\sqrt{2}/3)(u - (v + s)/2)$, and the edge length relatively by $(v + s)/2$. For b_x , the first-order wedge is $X \geq (1 - b_x)|Y|/\sqrt{2}$: the angle derivative is $(\sqrt{2}/3)b_x$ and the relative edge derivative is b_x . Reflection symmetry makes the other two off-diagonal angular derivatives zero. Translating the apex contributes $-A_{12}hz_x$ to its area derivative, with opposite sign on 34. These calculations give (15) and its cyclic versions.

Let $H(M) = 2\mathcal{C}(MM^T) - |M|_F^2$. Its score gradient is $2(B(M) - M)$. At the regular point $L = P$, so

$$H'' = 2 \sum_{i < j} \delta w_{ij} (\delta m_i - \delta m_j) \cdot (m_i^0 - m_j^0).$$

Write $(u, v, s) = \rho(1, 1, 1) + (u_0, v_0, s_0)$ with sum of the latter zero. Substitution of (15) and $z = h(b_x, b_y, b_z)$ gives

$$H'' = -24a^2\rho^2 - 2a^2(3c_0 + 1)(u_0^2 + v_0^2 + s_0^2) - 8a^2(1 + h^2 - c_0)(b_x^2 + b_y^2 + b_z^2). \quad (16)$$

The diagonal terms follow from $4a^2 \sum K_x(v + s)$; each off-diagonal pair contributes $-8a^2(1 + h^2 - c_0)b_x^2$. Also $0 < c_0 < 1$, since $\arctan \sqrt{2} > \sqrt{2}/3$ by the integral defining arctangent. All coefficients in (16) are strictly negative. Smoothness and Taylor's theorem give a strict local maximum of H modulo rotations.

For $Q \in \mathcal{K}$ near Q_* choose centered scores with Gram matrix Q and multiply them by c_* . They have squared norm F_* and are near the regular self-moment list modulo rotations. Since $H = 2c_*\mathcal{C}(Q) - F_*$, the preceding strict inequality proves the claimed strict local maximum on the covariance domain. $\square$

7. Perimeter bounds for full-rank critical values

We import the following precise case of the Gaussian multi-bubble theorem [MN22, Theorem 1.1]: among all four-cell partitions in $\mathbb{R}^3$ with masses $1/4$, the total Gaussian interface perimeter $S = \frac{1}{2} \sum_i P_\gamma(C_i) = \sum_{i < j} A_{ij}$ is at least its value for the regular tetrahedral fan. The theorem applies because the number of cells is at most dimension plus one. Its simplicial minimizer is central in this equal-mass case: for regular scores the balancing prices are unique modulo a constant by Lemma 3, and symmetry makes all prices equal. The regular value computed above is

$$S_* = 6\phi(0)\theta/\pi = c_*\sqrt{3/2}. \quad (17)$$

This is an established external theorem, not a new proof of the multi-bubble result. Its uniqueness part is not needed here.

Lemma 8. Any full-rank $Q \in \mathcal{K}$ at which the trace-constrained derivative of $\mathcal{C}$ vanishes satisfies $\mathcal{C}(Q) \geq c_*$.

Proof. By (6), $L = \mu P$ on $\mathbf{1}^\perp$, and by (5) $\mathcal{C}(Q) = \mu$. Each weight is $\mu/4$. Since the rows of M sum to zero and their squared norms sum to one,

$$\sum_{i < j} |v_i - v_j|^2 = 4, \quad S = \frac{\mu}{4} \sum_{i < j} |v_i - v_j| \leq \mu \sqrt{3/2}.$$

Combine this upper bound with (17) to get $\mu \geq c_*$. The perimeter theorem gives a *lower* critical-value bound; the next argument is necessary to turn it into a global *upper* bound. $\square$

8. A constrained mountain-pass lemma with the required sign

Lemma 9. Let K be nonempty compact convex in a finite-dimensional Euclidean space, and let f be C^2 on a neighborhood of K . Let $x_0, x_1 \in K$. Suppose every continuous path in K joining them meets a set on which $f \leq a < \min(f(x_0), f(x_1))$. Then some $x \in K$ has $f(x) \leq a$ and

$$\langle \nabla f(x), y - x \rangle \leq 0 \quad \text{for all } y \in K. \quad (18)$$

Proof. Let Γ be these paths and put $b = \sup_{\gamma \in \Gamma} \min_t f(\gamma(t)) \leq a$. The family is nonempty by convexity. Let Π_K be Euclidean projection and $D(x) = \Pi_K(x + \nabla f(x)) - x$. The projection inequality tested at $x \in K$ gives $\nabla f(x) \cdot D(x) \geq |D(x)|^2$. Moreover $D(x) = 0$ exactly when (18) holds. This vector field is locally Lipschitz, since f is C^2 and projection is nonexpansive.

If the level b contains no zero of D , compactness gives $\eta, \alpha > 0$ with $|D| \geq \alpha$ on $\{|f - b| \leq 2\eta\} \cap K$. The level is nonempty by any path and the intermediate value theorem. Shrink η so both endpoint values exceed $b + 2\eta$. Choose a Lipschitz cutoff $0 \leq \chi \leq 1$, one for $|f - b| \leq \eta$ and zero for $|f - b| \geq 2\eta$. The flow $x' = \chi(f(x))D(x)$ stays in K : every sufficiently small forward Euler step is a convex combination of x and $\Pi_K(x + \nabla f(x))$. Euler approximations converge to the locally Lipschitz flow, giving invariance. Compactness permits any finite forward time and continuous dependence on the initial point. Along the flow f never decreases and increases at rate at least α^2 while $|f - b| \leq \eta$; the endpoints are fixed.

Choose γ with minimum greater than $b - \eta/2$. Applying a fixed flow time greater than η/α^2 makes every point on this path have value at least $b + \eta/2$. Points already above this value do not descend, and the others reach it within that time. This continuous path contradicts the definition of b . Thus a zero exists at level b , proving the lemma. $\square$

The sign in (18) is the upper normal cone, which is essential below. The proof uses only Euclidean projection and the elementary existence, Euler approximation and continuous-dependence theorems for locally Lipschitz finite-dimensional ODEs. It does not assume nondegenerate critical points or a smooth boundary of K .

9. Regularization excludes every additional maximizer

Theorem 10 (covariance form). For $Q \in \mathcal{K}$, $\mathcal{C}(Q) \leq c_*$, with equality only at Q_* .

Proof. Suppose $Q_1 \neq Q_*$ has $\mathcal{C}(Q_1) \geq c_*$. By Lemma 7 choose $r, \delta > 0$ so that the closed radius- r ball about Q_* within the trace affine space lies in the positive cone, Q_1 is outside it, and its boundary has $\mathcal{C} \leq c_* - \delta$. For $\varepsilon \downarrow 0$ define on $\mathcal{K}$

$$\mathcal{C}_\varepsilon(Q) = \mathcal{C}(\widetilde{Q}), \quad \widetilde{Q} = (1 - \varepsilon)Q + \varepsilon Q_*.$$

This is smooth on an open neighborhood of $\mathcal{K}$, and converges uniformly to $\mathcal{C}$ by Lemma 2 and compactness. For small ε , its values at Q_* , Q_1 exceed $c_* - \delta/3$, whereas every joining path meets the sphere with value at most $c_* - 2\delta/3$. Lemma 9 gives $Q_\varepsilon \in \mathcal{K}$ satisfying the upper-normal condition and

$$\mathcal{C}(\widetilde{Q}_\varepsilon) \leq c_* - 2\delta/3. \quad (19)$$

Let L_ε be its full-rank facet matrix. By (6), the normal condition says that Q_ε maximizes $\text{tr}(L_\varepsilon Q)$ on $\mathcal{K}$. If μ_ε is the largest eigenvalue of L_ε on $\mathbf{1}^\perp$, this is equivalent to

$$0 \preceq L_\varepsilon \preceq \mu_\varepsilon P, \quad (L_\varepsilon - \mu_\varepsilon P)Q_\varepsilon = 0. \quad (20)$$

Indeed the linear functional on positive trace-one matrices is at most its top eigenvalue. Equality makes the range of Q_ε lie in its top eigenspace; this follows by diagonalizing L_ε and using nonnegative diagonal entries of Q_ε (a zero diagonal in a positive matrix has zero row).

Choose once and for all a 4×3 isometry U onto $\mathbf{1}^\perp$ and set $M_\varepsilon = \widetilde{Q}_\varepsilon^{1/2}U$, $B_\varepsilon = L_\varepsilon M_\varepsilon$. Equations (5) and (20) imply

$$\begin{aligned} \mathcal{C}(\widetilde{Q}_\varepsilon) &= (1 - \varepsilon)\mu_\varepsilon + \varepsilon \text{tr } L_\varepsilon / 3, \\ 0 < \mu_\varepsilon &\leq 2h_0/(1 - \varepsilon), \\ |B_\varepsilon - \mu_\varepsilon M_\varepsilon|_F^2 &= \frac{\varepsilon}{3} \text{tr} [(L_\varepsilon - \mu_\varepsilon P)^2] \leq \varepsilon \mu_\varepsilon^2. \end{aligned} \quad (21)$$

For the last identity, insert $M_\varepsilon M_\varepsilon^T = (1 - \varepsilon)Q_\varepsilon + \varepsilon P/3$ into the squared norm; the Q_ε term vanishes by (20). The three eigenvalues of L_ε lie in $[0, \mu_\varepsilon]$, giving the inequality.

Lemma 4 and (21) yield, for every pair,

$$k_0 \leq \mu_\varepsilon (|v_i^\varepsilon - v_j^\varepsilon| + \sqrt{2\varepsilon}). \quad (22)$$

The factor $\sqrt{2}$ bounds a difference of two rows of the residual by $\sqrt{2}$ times its Frobenius norm. Since $|M_\varepsilon|_F^2 = 1$, all pair distances are at most $\sqrt{2}$. Thus (22) also bounds μ_ε away from zero. Together with its uniform upper bound, it keeps all score pairs uniformly separated as $\varepsilon \rightarrow 0$.

Pass to a subsequence with $Q_\varepsilon \rightarrow Q_0 \in \mathcal{K}$, $\mu_\varepsilon \rightarrow \mu_0 > 0$ and $M_\varepsilon \rightarrow M_0$. Normalize prices by minimum zero. Equation (10) bounds them, so take a further convergent subsequence. Lemma 5 gives balanced limiting cells and $L_\varepsilon \rightarrow L_0$. Their moments satisfy

$$B_0 = L_0 M_0 = \mu_0 M_0, \quad L_0 \preceq \mu_0 P. \quad (23)$$

If Q_0 has rank at most two, multiply its scores and prices by μ_0 . The winning cells do not change; their inducing vectors become their own moments by (23), and their facet matrix becomes $L_0/\mu_0 \preceq$

P . Their rows remain distinct, contradicting Lemma 6. If Q_0 has rank three, (23) gives $L_0 = \mu_0 P$, so Lemma 8 gives $\mathcal{C}(Q_0) \geq c_*$. But continuity and (19) give $\mathcal{C}(Q_0) \leq c_* - 2\delta/3$. This is again a contradiction. There is no Q_1 as assumed, proving the theorem. $\square$

10. Completion of the partition theorem

Let M have rows m_i for an arbitrary partition in Theorem 1. Its rows sum to zero, so its rank is at most three, and $\text{tr}(MM^T) = F$. For every price, its labels are a feasible linear assignment in (2), giving

$$F \leq \mathcal{C}(MM^T).$$

For $F > 0$, Theorem 10 and homogeneity imply $F \leq c_* \sqrt{F}$, which is (1). The case $F = 0$ is strictly smaller. Equality requires $MM^T = F_* P/3$, by Theorem 10, and equality in the pointwise optimal assignment. These four regular scores have null ties; their equal-mass prices are all equal by symmetry and uniqueness. Hence the original partition is the regular winning partition. Fractional partitions have the identical feasible-assignment inequality; equality likewise forces their labels to be the unique winning label almost everywhere. This proves the stated extension and classification.

As a further consequence, every full-rank balanced self-moment stationary diagram is regular: its normalized score covariance is an interior critical point, so Lemma 8 puts its value at least c_* and Theorem 10 puts it at most c_* with a unique attaining covariance. This is not an assertion that lower-rank stationary saddles do not exist.

11. Dependencies and verification scope

The analytic inputs are the classical single-cell Gaussian isoperimetric theorem, the Gaussian multi-bubble perimeter theorem [MN22], and elementary finite-dimensional analysis (dominated convergence, implicit functions, polar decomposition, spectral decomposition, Gaussian integration by parts, and locally Lipschitz ODE flow). The three-cell estimate, profile margin, local tetrahedral calculation, constrained deformation and limiting-rank obstructions have been proved above.

No floating data, optimized solver, numerical root, or unverified quadrature sign is a premise of Theorem 1. The older covariance concavity route is unnecessary; this theorem does not prove that stronger concavity statement. The partial formal checks in this package explicitly state their scalar hypotheses and do not formalize Gaussian integrals, perimeter, the ODE deformation or the complete theorem. Internal model review is documented separately and is not external human peer review.

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